HDOC Day-59

Question: Write a C program that passes a string by address to longest_unique_substring(), finds the longest substring without repeated characters, and displays its length. Uppercase and lowercase characters are different; empty input gives 0.

C Program

```
#include <stdio.h>
#include <string.h>

void longest_unique_substring(const char *s) {
    int last[256];
    int start = 0, maxLen = 0, bestStart = 0;

    for (int i = 0; i < 256; i++)
        last[i] = -1;

    for (int i = 0; s[i] != '\0'; i++) {
        unsigned char ch = (unsigned char)s[i];
        if (last[ch] >= start)
            start = last[ch] + 1;
        last[ch] = i;

        if (i - start + 1 > maxLen) {
            maxLen = i - start + 1;
            bestStart = start;
        }
    }

    printf("Length of longest substring: %d\n", maxLen);
    printf("Longest substring: %.*s\n",
           maxLen, s + bestStart);
}

int main(void) {
    char s[1000];
    printf("Enter a string: ");
    if (fgets(s, sizeof(s), stdin) == NULL)
        return 0;
    s[strcspn(s, "\n")] = '\0';
    longest_unique_substring(s);
    return 0;
}
```

Sample Output

```
Enter a string: abcabcbb
Length of longest substring: 3
Longest substring: abc
```

Explanation: The function receives the string using a pointer. It tracks the last position of each character and moves the start of the current substring when a character repeats. It prints the maximum length and the corresponding substring. The character lookup is case-sensitive. For empty input, the maximum length stays 0.